STINT 2026 - LEUVEN, BELGIUM

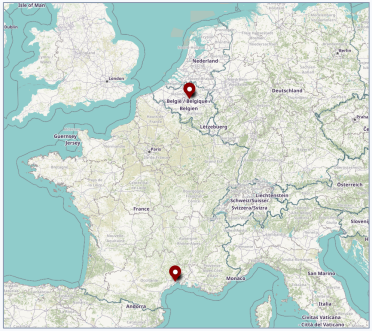
CSPCL

A stepwise, non-disruptive transition to Delay-Tolerant Networking for CubeSat operators.

Mathias Durat
Sarah Theoulle
Hugo Ponthieu
Tristan-Mihai Radulescu
Olivier De Jonckère

LIRMM · Polytech Montpellier · CSUM

Meet CSUM





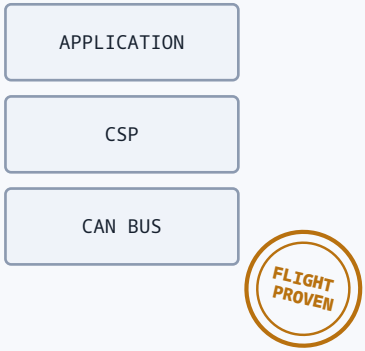
Montpellier, France

The control room



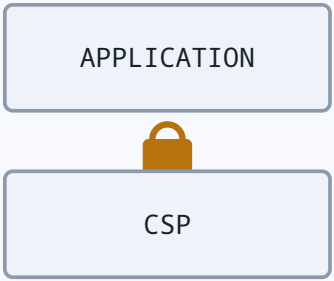
How to go DTN, non-intrusively?

CSP over CAN



"It just works."

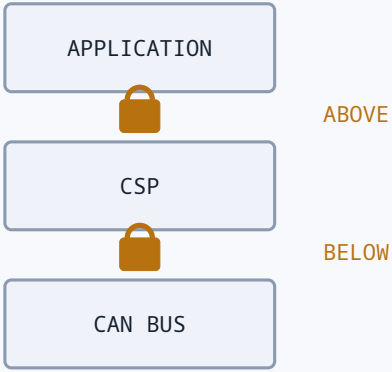
One small issue



ABOVE

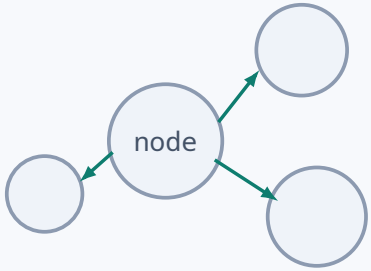
CSP must remain, above.

Two small issues



CSP must remain, above and below.

The gap



DTN network

----- ----- no path today

CSP-ONLY APP

Why above

- **Zero rewrite**
- **Non-intrusive**

Why below

- **Real hardware**
- **Portable**
- Small teams. Small budgets.

The solution



The CSP-BP sandwich.

Space agencies are already converging on BP

IOAG | NASA LCRNS | ESA Moonlight | NASA--ESA LunaNet

Charon



A tunnel: CSP, over Bundle Protocol.

Why Charon



rewritten, or no DTN



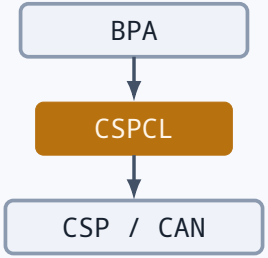
unchanged, and DTN

How Charon works



LEVEL 1 REACHED

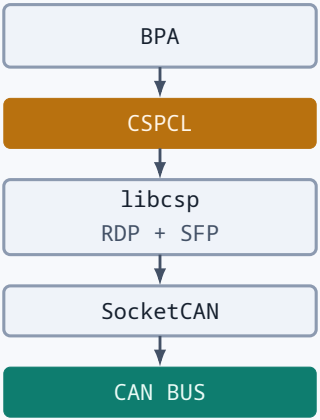
CSPCL



A Convergence Layer Adapter: bundles, over CSP.

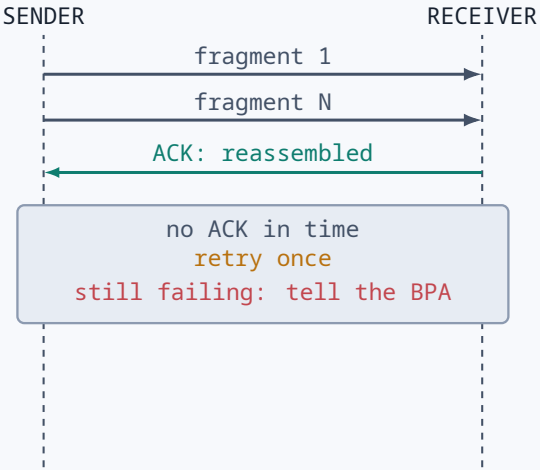
BP needs a ride.
CSP is the road.

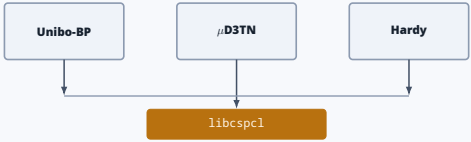
How CSPCL works



LEVEL 2 REACHED

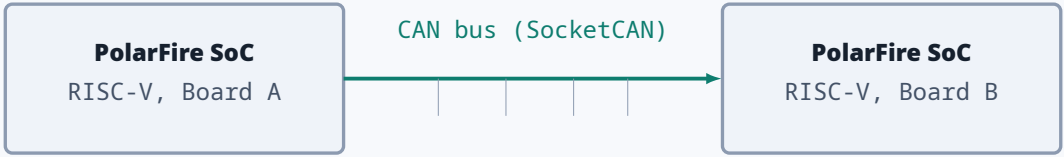
CSPCL — reliability & portability

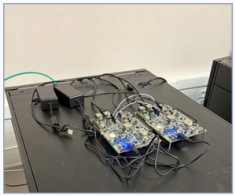




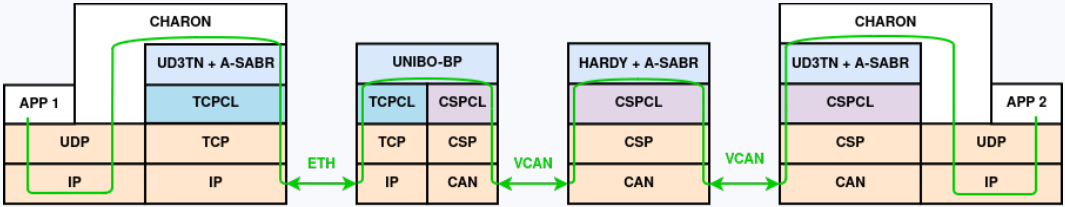
LEVEL 3 REACHED

The testbed

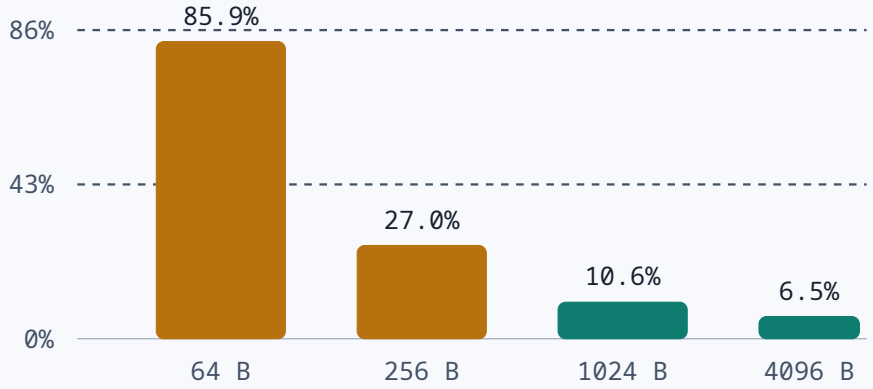

PolarFire SoC
RISC-V, Board A
CAN bus (SocketCAN)
PolarFire SoC
RISC-V, Board B



The full chain

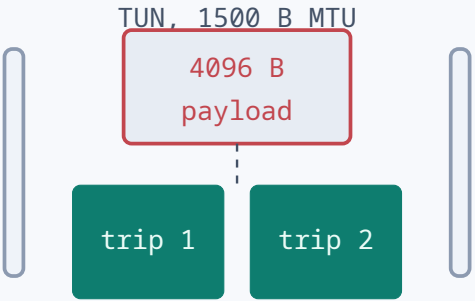


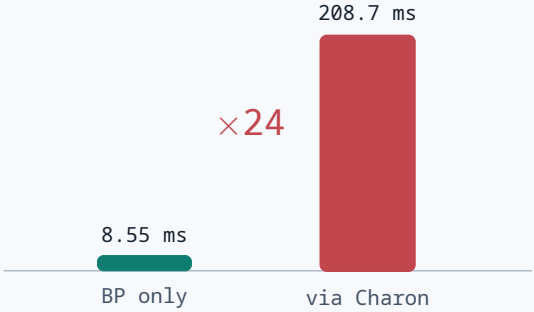
Framing overhead



Batch telemetry into bigger bundles.

An MTU surprise





Throughput & delivery



78.9% and 94.1% are throughput as % of the contact plan’s modeled ceiling

100% delivery

over 100 iterations

Disruption test

```
$ kill -9 unibo-bp-cspcl
[bundle] held in storage
[unibo-bp-cspcl] restarted
[bundle] delivered
```

Store-and-forward, validated.

Charon — above

- **Single BPA support**
- **No local packet queue**
- **No CSP routing in CSP mode**
- **Next: beyond μD3TN**

CSPCL — below

- **Use A-SABR for routing**
- **Next: micro-controller support**
- **Next: narrow the pool mutex**

THANK YOU

Questions?

Non-disruptive DTN adoption for CSP-based CubeSats.

Not just a paper idea.

Mathias Durat · Sarah Theoulle · Hugo Ponthieu · Tristan-Mihai Radulescu · Olivier De Jonckère

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